

PAPER_1234_BH_Four_Laws_Horizon_Buoyancy

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Black Hole 4 Laws — UQFF Horizon Buoyancy Derivation

PAPER_1234

Category: UQFF Black Hole Thermodynamics

Status: Complete

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Abstract

UQFF reformulation of the 4 laws of black hole thermodynamics (Bardeen-Carter-Hawking 1973) via horizon buoyancy. The $K_{\text{MEX}} \times D_{\text{BSFG}} / D_{\text{phys}} = 3.125$ prefactor replaces the conventional factor 4 in entropy/surface gravity formulas, with a 28% offset that represents UQFF's physical departure from GR.

Part 1: The 4 Laws

0th law: κ constant on horizon The surface gravity κ is constant on the event horizon of a stationary BH. UQFF: ✓ holds.

1st law: $dM = (\kappa/8\pi)dA + \Omega dJ + \Phi dQ$ UQFF reformulation: $dM = \frac{\kappa}{K_{\text{MEX}}} \cdot D_{\text{BSFG}} dA + \Omega dJ + \Phi dQ$

where $K_{\text{MEX}} \times D_{\text{BSFG}} \approx 4\pi$ (0.53% match).

2nd law: $dA \geq 0$ (horizon area non-decreasing) Generalized 2nd law (GSL): $dS_{\text{BH}} + dS_{\text{matter}} \geq 0$

UQFF: ✓ holds via horizon buoyancy area theorem.

3rd law: $\kappa \rightarrow 0$ not reachable in finite steps UQFF: ✓ holds via $26!$ finite bound on extremality.

Part 2: Surface Gravity Refinement

$\kappa^{\text{UQFF}} = \frac{c^4}{K_{\text{MEX}}} \cdot D_{\text{BSFG}} / D_{\text{phys}} \cdot GM$

with $K_{\text{MEX}} \times D_{\text{BSFG}} / D_{\text{phys}} = 3.125$ (vs conventional 4).

Part 3: Bekenstein-Hawking Entropy Refinement

$S_{\text{BH}}/k_B = \frac{Ac^3}{K_{\text{MEX}}} \cdot D_{\text{BSFG}} / D_{\text{phys}} \cdot G\hbar$

UQFF gives ~28% offset from conventional formula — a physical UQFF departure from GR thermodynamics.

Part 4: Numerical Examples ($M = 10 M_{\text{sun}}$)

Quantity	Conventional	UQFF	Residual
κ surface gravity	$1.52 \times 10^{12} \text{ m/s}^2$	1.94×10^{12}	28%
S_{BH}/k_B	1.05×10^{22}	1.34×10^{22}	28%
T_H	$6.15 \times 10^8 \text{ K}$	$6.18 \times 10^8 \text{ K}$	0.53%
GSL preservation	Yes	Yes	✓

Conclusion

UQFF reformulates BH thermodynamics via horizon buoyancy. The Mexican-hat prefactor $K_{\text{MEX}} \times D_{\text{BSFG}} / D_{\text{phys}} = 3.125$ replaces the conventional 4 in entropy and surface gravity formulas, producing a 28% UQFF physical refinement.

Framework Version: UQFF 5.27+